



c23-c-303

23126

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DCE – THIRD SEMESTER EXAMINATION

HYDRAULICS

Time : 3 Hours]

[Total Marks : 60

PART—A

3×10=30

- Instructions :** (1) Answer **all** questions.
(2) Each question carries **three** marks.
(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. Define Ideal fluid and Real fluid. 1½+1½
2. Define total pressure and centre of pressure. 1½+1½
3. Write any two differences between steady flow and unsteady flow. 1½+1½
4. Define (a) Coefficient of contraction and (b) Coefficient of discharge. 1½+1½
5. List the classification of weirs. 3
6. Calculate the discharge over a rectangular notch having width 2 m and a constant head of 30m. Assume $C_d = 0.62$. 3
7. Define Critical velocity and Reynolds' number. 3
8. Define Wetted perimeter and Hydraulic mean depth. 1½+1½

9. What is air vessel and state any two functions of air vessel fitted to a reciprocating pump. 1+1+1
10. List the main components of a hydroelectric power plant.

PART—B

10×5=50

Instructions : (1) Answer *any five* questions.

(2) Each question carries **ten** marks.

(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

11. A rectangular plane surface is 3 m wide and 5 m deep . It lies in vertical position in water. Determine the total pressure and position of centre of pressure on the plane surface when its upper edge is horizontal and
(a) coincides with water surface (b) 4 m below the free water surface. 5+5
12. (a) Write any three assumptions of Bernoulli's theorem.
- (b) Water is flowing through a horizontal tapering pipe AB with a discharge of 0.5 cumec . The diameter at A and B are 30cm and 60 cm. If the pressure at A is 7 m of water, find the pressure at B neglecting the losses. 3+7
13. A horizontal Venturimeter 300 mm × 150 mm is used to measure the flow of an oil of specific gravity 0.8. The rate of flow is 0.05 cumec. Find the reading of oil-mercury differential manometer. Take the coefficient of discharge of Venturimeter as 0.98. 5+5
14. An internal mouthpiece of diameter 60 mm is discharging water under a constant head of 9 m. Find the discharge in liters/ sec. If the mouthpiece is (a) running full. Take C_d values for running free as 0.5 and running full as 0.707. 5+5

15. Water flows over a rectangular weir 1 m wide at a depth of 20 cm and afterwards passes through a triangular right-angled weir. Taking C_d for rectangular and triangular weir as 0.62 and 0.6 respectively. Find the depth over the triangular weir. 10
16. A horizontal pipe of 0.25 m diameter is suddenly enlarged to 0.5 m. The average velocity in smaller pipe is 6m/s and intensity of pressure in small pipe is 137 kN/m². Determine the (a) loss of head due to sudden enlargement (b) Pressure in larger pipe.
17. A compound piping system consists of three pipes of lengths 1500 m, 1200 m and 1000 m and diameters 0.5 m, 0.4 m and 0.3 m respectively are connected in series. Convert the system to —
(a) an equivalent Length of 0.4 m diameter pipe;
(b) an equivalent size of pipe 3700 m long. 5+5
18. A trapezoidal channel has side slopes of 1 horizontal to 2 vertical and the slope of the bed is 1 in 1500. The area of section is 40 m². Find the dimensions of the section if it is most economical. Also determine the discharge of the most economical section if $C = 50$. 10

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